Lab 04 - Counting Sort (Part 2)

Instruction:

- The objective of this part (Part 2) is to define a variation of the counting sort that operates words (strings of letters) while maintaining linear running time. The procedures of this lab are to:
  - Define a hash function that maps letters to integers.
  - Define a variation of the counting sort.
  - Derive the running-time function of the variation of the counting sort and its helper functions.
  - Test the variation of the counting sort on a randomly generated integer array.
- Submit the modified `.h` file from Part 1, one `.txt` file, and one `.cpp` source file.
- Your files may include only the following libraries: `iostream`, `sstream`, `string`, `vector`, `ctime`, `cctype`, and `cstdlib`. No others are permitted.
- Submit your files to your GitHub repository, inside a directory named `Lab04`.
- Cheating of any kind is prohibited and will not be tolerated.
- **Violating or failing to follow any of these rules will result in an automatic zero (0) for the lab.**

At the very beginning of your submission, add a comment containing your name and the date, to confirm you have read and understood the instructions above.

Grading

Task	Maximum Points	Points Earned
1	0.5	
2	0.5	
3	0.5	
4	0.5	
5	0.5	
Total	2.5	

Task 1

In the header file, within the namespace *aal*, define the function:

```
size_t hash(char c)
```

that returns the zero-based position of *c* in the alphabet (that is, its position in alphabetical order minus 1) if *c* is a letter; otherwise, it returns 26.

Task 2

In the header file, within the namespace *aal*, define the function:

```
bool isWord(const string& str)
```

that returns true if *str* contains only letters; otherwise, it returns false.

Task 3

In the header file, within the namespace *aal*, define the function:

```
string Anagram(const string& A)
```

that returns a sorted loowercase copy of *A*, obtained by using the counting sort algorithm with the `hash()` function, if *A* is a word; otherwise, it returns a copy of *A*.

Task 4

In the text file, construct the running-time table of `hash()`, `isWord()`, and `Anagram()`, and derive their running-time functions. Assume that each statement has a cost of 1. Show all of your work.

Task 5

In the `main` function, verify that your `Anagram()` function is correct by completing the following steps:

1. Declare two string objects.
2. Prompt the user to enter two words, and store the inputs in the objects.
3. Display each object.
4. Compare the results of invoking `Anagram()` on each object. If the results are equal, display the message `"They are anagrams of each other"`; otherwise, display the message `"They are not anagrams of each other"`.
5. Repeat steps 2 through 4, until the inputs are anagrams.